Resolution Algorithm – Complete Solutions

Question 1 – Rain and Wet Ground

Let R = It is raining and W = The ground is wet.

Propositional logic:

$R \rightarrow W$

R

Goal: W

CNF:

C1: $\neg R \vee W$

C2: R

Negated goal:

C3: $\neg W$

Resolution:

C1 ($\neg R \vee W$) and C3 ($\neg W$) resolve to C4: $\neg R$.

C2 (R) and C4 ($\neg R$) resolve to C5: ■.

Conclusion: The Empty Clause is derived, so the goal is proved. Therefore, the ground is wet.

Question 2 – Student Assignment Submission

Let S = Rahul submitted the assignment and M = Rahul receives internal marks.

Propositional logic:

$S \rightarrow M$

S

Goal: M

CNF:

C1: $\neg S \vee M$

C2: S

Negated goal:

C3: $\neg M$

Resolution:

C1 (¬S ∨ M) and C3 (¬M) resolve to C4: ¬S.

C2 (S) and C4 (¬S) resolve to C5: ■.

Conclusion: The Empty Clause is derived. Therefore, Rahul receives internal marks.

Question 3 – Library Membership

Let L = Priya is a library member and B = Priya can borrow books.

Propositional logic:

L → B

L

Goal: B

CNF:

C1: ¬L ∨ B

C2: L

Negated goal:

C3: ¬B

Resolution:

C1 (¬L ∨ B) and C3 (¬B) resolve to C4: ¬L.

C2 (L) and C4 (¬L) resolve to C5: ■.

Conclusion: The Empty Clause is derived. Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

Let A = Arun cleared the aptitude test and E = Arun is eligible for placement.

Propositional logic:

A → E

A

Goal: E

CNF:

C1: ¬A ∨ E

C2: A

Negated goal:

C3: ¬E

Resolution:

C1 (¬A ∨ E) and C3 (¬E) resolve to C4: ¬A.

C2 (A) and C4 (¬A) resolve to C5: ■.

Resolution tree:

¬A ∨ E ¬E

\ /

¬A

|

A

|

■

Conclusion: The Empty Clause is derived. Therefore, Arun is eligible for placement.

Question 5 – Access Control System

Let P = User entered the correct password, A = User is authenticated, and G = User is granted access.

Propositional logic:

P → A

A → G

P

Goal: G

CNF:

C1: ¬P ∨ A

C2: ¬A ∨ G

C3: P

Negated goal:

C4: ¬G

Resolution:

Step 1: C2 (¬A ∨ G) and C4 (¬G) resolve to C5: ¬A.

Step 2: C1 (¬P ∨ A) and C5 (¬A) resolve to C6: ¬P.

Step 3: C3 (P) and C6 (¬P) resolve to C7: ■.

Resolution tree:

¬A ∨ G ¬G

\ /

¬A

|

¬P ∨ A

|

¬P

|

P

|

■

Conclusion: The Empty Clause is derived. Therefore, the user is granted access.